

Linear Algebra — GATE DA Revision Notes

1. Vector Space

1. Big Picture A vector space is a set where "addition" and "scalar multiplication" behave nicely — closure, associativity, existence of zero/inverse, distributivity. Everything else (basis, dimension, rank) is built on top of this skeleton.

2. Intuition Any collection of objects (vectors, polynomials, matrices, functions) that you can add together and scale, and still stay inside the same collection, forms a vector space. Think of it as "a playground closed under +ve combinations."

3. Formal Definition A set V over a field F (usually $\mathbb{R}$) is a vector space if for all $u, v, w \in V$ and $a, b \in F$:

- $u + v \in V$ (closure under addition)
- $u + v = v + u, (u + v) + w = u + (v + w)$
- $\exists 0 \in V: u + 0 = u$
- $\exists -u \in V: u + (-u) = 0$
- $au \in V$ (closure under scalar mult.)
- $a(u + v) = au + av, (a + b)u = au + bu, a(bu) = (ab)u, 1 \cdot u = u$

4. Explanation of Terms

- **Field F :** the scalars (e.g. $\mathbb{R}, \mathbb{C}$) used to scale vectors.
- **Closure:** the operation never takes you outside the set.
- **Zero vector:** additive identity, unique in every vector space.

5. Important Properties

- Zero vector is unique — *if two zeros existed, adding one to other gives both equal.*
- Additive inverse of each vector is unique — *follows from cancellation using associativity.*
- $0 \cdot u = 0$ (scalar zero times vector = zero vector) — **distributes as $0u + 0u = 0u \Rightarrow 0u = 0$.**
- $a \cdot 0 = 0$ — *similarly, scalar times zero vector is zero vector.*

6. Theorems GATE Asks

- Checking "is this a vector space?" reduces to checking **closure under addition and scalar multiplication** (rest usually inherited) — apply when a subset of $\mathbb{R}^n$ or function space is given.
- Any vector space has a **basis**, and all bases have the **same size = dimension** — apply when asked to find dimension of an abstract space (polynomials, matrices, etc.).

7. Important Formulae

- $\dim(\mathbb{R}^n) = n$
- $\dim(P_n(x)) = n + 1$ (polynomials of degree $\leq n$)
- $\dim(M_{m \times n}) = mn$

8. Geometric Interpretation $\mathbb{R}^2, \mathbb{R}^3$ are the visual playgrounds: vectors as arrows from origin, addition = tip-to-tail, scaling = stretching/shrinking/flipping.

9. Common Examples

- $\mathbb{R}^n$, set of all $m \times n$ matrices, set of polynomials of degree $\leq n$, set of continuous functions on $[a, b]$.

10. Common Non-Examples

- Set of vectors in $\mathbb{R}^2$ with **non-negative** coordinates (fails closure under scalar mult. by negative numbers).
- Set of solutions to $Ax = b$ with $b \neq 0$ (no zero vector, not closed under addition).

11. Common Mistakes Students Make

- Assuming any subset of $\mathbb{R}^n$ containing origin is automatically a vector space (still need closure under both operations).
- Forgetting to check scalar multiplication closure (only checking vector addition).

12. GATE Traps

- A set defined by an equation like $x + y + z = 1$ (affine, not through origin) is NOT a vector space — no zero vector.
- Union of two subspaces is generally **not** a subspace (only intersection is guaranteed).

13. Memory Tricks "CAZI-D" — Closure, Associativity, Zero, Inverse, Distributive — the 5 pillars to check.

14. One-Page Summary Vector space = set closed under addition & scalar multiplication + 8 axioms. Test: pick two elements, check sum stays in set; pick one element, check scalar multiple stays in set; check zero vector exists. $\mathbb{R}^n$, matrices, polynomials are classic examples; affine sets (not through origin) are classic non-examples.

2. Subspaces

1. Big Picture A subspace is a "vector space living inside a bigger vector space." It's the main tool to describe solution sets, column spaces, null spaces — anything that stays closed under linear combinations.

2. Intuition Take any subset of a vector space that itself doesn't "leak out" when you add or scale its elements — that's a subspace. It must pass through the origin.

3. Formal Definition A subset $W \subseteq V$ is a subspace of V if:

1. $0 \in W$

2. $u, v \in W \Rightarrow u + v \in W$
3. $u \in W, a \in F \Rightarrow au \in W$

(Equivalently: W is closed under linear combinations, $au + bv \in W$.)

4. Explanation of Terms

- **Zero vector condition:** quick first check — if $0 \notin W$, stop, not a subspace.
- **Closed under linear combination:** single combined check for addition + scalar multiplication.

5. Important Properties

- Intersection of subspaces is a subspace — *closure conditions hold in both simultaneously, hence in the intersection.*
- Sum of subspaces $W_1 + W_2 = \{w_1 + w_2\}$ is a subspace — *closed by construction.*
- Union of subspaces is generally NOT a subspace — *sum of one vector from each may leave both.*
- Every subspace contains the zero vector — *direct from definition.*

6. Theorems GATE Asks

- **Dimension formula:** $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$ — apply when asked for dimension of sum of two given subspaces.
- Column space, row space, null space of a matrix are all subspaces — apply when checking properties of $Ax = 0$ or $Ax = b$ solution sets.

7. Important Formulae

- $\dim(W_1 \cap W_2) = \dim W_1 + \dim W_2 - \dim(W_1 + W_2)$
- Null space dimension = nullity = $n - \text{rank}(A)$

8. Geometric Interpretation In $\mathbb{R}^3$: subspaces are only the origin, lines through origin, planes through origin, or all of $\mathbb{R}^3$ (dimension 0,1,2,3).

9. Common Examples

- Null space $N(A)$, column space $C(A)$, row space of a matrix.
- Any line/plane through the origin.

10. Common Non-Examples

- A line not passing through origin.
- Set of invertible matrices (not closed under addition; sum of two invertible matrices can be singular).

11. Common Mistakes Students Make

- Checking only that $0 \in W$ and stopping (must verify closure too).
- Believing union of two subspaces is a subspace.

12. GATE Traps

- Given two planes through origin in $\mathbb{R}^3$, their union is NOT a subspace even though both individually are.
- Solution set of $Ax = b$ ($b \neq 0$) is an affine set, not a subspace — only $Ax = 0$ gives a subspace.

13. Memory Tricks "Subspace must hug the origin and never escape under +/scale."

14. One-Page Summary Subspace $W \subseteq V$: contains 0, closed under addition and scalar multiplication. Intersection of subspaces = subspace; union generally isn't. Dimension formula: $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$. Column/row/null space of a matrix are the GATE-favorite subspace examples.

3. Linear Dependence and Independence of Vectors

1. Big Picture Linear independence tells you whether a set of vectors carries "redundant" information. It is the backbone concept behind basis, rank, and dimension.

2. Intuition Vectors are dependent if at least one of them can be written as a combination of the others (redundant direction). Independent means every vector adds a genuinely new direction.

3. Formal Definition Vectors $v_1, \dots, v_k \in V$ are **linearly independent** if the only solution to

$$c_1v_1 + c_2v_2 + \dots + c_kv_k = 0$$

is $c_1 = c_2 = \dots = c_k = 0$. If a nontrivial solution exists, they are **linearly dependent**.

4. Explanation of Terms

- **Trivial solution:** all scalars $c_i = 0$ — always satisfies the equation.
- **Nontrivial solution:** at least one $c_i \neq 0$ — indicates dependence.

5. Important Properties

- Any set containing the zero vector is automatically dependent — *choose $c = 1$ for zero vector, rest 0, gives nontrivial combination = 0.*
- More vectors than dimension $\Rightarrow$ automatically dependent — *cannot have more than n independent vectors in $\mathbb{R}^n$.*
- Independent set stays independent after removing vectors; dependent set stays dependent after adding vectors.
- Columns of A are independent $\iff Ax = 0$ has only trivial solution $\iff \text{rank}(A) = n$ (number of columns).

6. Theorems GATE Asks

- **Rank test:** vectors (as columns of matrix A) are independent iff $\text{rank}(A) = \text{number of vectors}$ — apply whenever asked to check independence of a given vector list; form matrix, row reduce.
- **Determinant test** (square case only): n vectors in $\mathbb{R}^n$ independent iff $\det(A) \neq 0$ — apply only when number of vectors = dimension.
- **Wronskian** (for functions): nonzero Wronskian at a point $\Rightarrow$ functions independent (converse not always true).

7. Important Formulae

- $\text{rank}(A) = \text{number of vectors} \Rightarrow$ independent.
- For square A : independent $\iff \det(A) \neq 0 \iff$ invertible.

8. Geometric Interpretation Dependent vectors in $\mathbb{R}^3$ lie in a common plane or line (lower-dimensional); independent vectors truly span more directions/volume.

9. Common Examples

- Standard basis vectors $e_1, \dots, e_n$ — always independent.
- Any set of orthogonal nonzero vectors — always independent.

10. Common Non-Examples

- $(1, 2), (2, 4)$ — scalar multiples, dependent.
- Any set including the zero vector.

11. Common Mistakes Students Make

- Checking independence by eyeballing instead of solving $Ax = 0$ / computing rank.
- Confusing "independent" with "orthogonal" (orthogonal $\Rightarrow$ independent, but not conversely required).

12. GATE Traps

- k vectors in $\mathbb{R}^n$ with $k > n$: dependent guaranteed — no need to compute anything.
- Non-square matrix: can't use determinant test — must use rank.

13. Memory Tricks "Only zero kills them all" — independence means only the trivial combination gives zero.

14. One-Page Summary Independent $\iff$ only trivial linear combination gives 0 $\iff$ rank = number of vectors $\iff$ (square case) $\det \neq 0$. More vectors than dimension $\Rightarrow$ automatically dependent. Zero vector in the set $\Rightarrow$ automatically dependent.

4. Matrices

1. Big Picture A matrix is a compact way to represent a linear transformation or a system of linear equations. Almost every other topic (rank, eigenvalues, decomposition) is a "property of a matrix."

2. Intuition Think of a matrix as a machine: feed in a vector x , it outputs Ax — a new vector, obtained via linear combination of A 's columns using x 's entries as weights.

3. Formal Definition An $m \times n$ matrix $A = [a_{ij}]$ is a rectangular array of scalars with m rows, n columns, representing a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m, T(x) = Ax$.

4. Explanation of Terms

- a_{ij} : entry in row i , column j .
- **Square matrix**: $m = n$.
- ****Transpose A^T ****: rows and columns swapped, $(A^T)_{ij} = a_{ji}$.
- **Symmetric matrix**: $A = A^T$.
- **Trace**: sum of diagonal entries, $\text{tr}(A) = \sum a_{ii}$.

5. Important Properties

- $(AB)^T = B^T A^T$ — *transpose reverses order because $(AB)_{ij} = \sum a_{ik} b_{kj}$ flips indices under transpose.*
- $\text{tr}(AB) = \text{tr}(BA)$ — *sum over diagonal is invariant to cyclic reordering (even though $AB \neq BA$ generally).*
- $(A^{-1})^{-1} = A, (AB)^{-1} = B^{-1} A^{-1}$ — *inverse also reverses order, mirroring transpose.*
- Matrix multiplication is associative but NOT commutative in general — *dimension mismatch or different linear map composition order.*

6. Theorems GATE Asks

- **Invertibility equivalences**: A invertible $\iff \det(A) \neq 0 \iff \text{rank} = n \iff$ columns independent $\iff Ax = 0$ has only trivial solution — apply whenever a "which of these is equivalent to invertible" question appears.
- **Rank-Nullity Theorem**: $\text{rank}(A) + \text{nullity}(A) = n$ (number of columns) — apply for any $m \times n$ matrix.

7. Important Formulae

- $\det(AB) = \det(A) \det(B)$
- $\det(A^T) = \det(A)$
- $\det(A^{-1}) = 1 / \det(A)$
- $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$

8. Geometric Interpretation A matrix transforms space: rotates, scales, shears, or projects. Determinant = volume scaling factor (with sign for orientation flip).

9. Common Examples

- Identity matrix I (does nothing), diagonal matrices (pure scaling along axes), rotation matrices.

10. Common Non-Examples

- A non-square array is not a "linear operator on itself" (maps between different-dimension spaces) — still a matrix, just not square.

11. Common Mistakes Students Make

- Assuming $AB = BA$.
- Forgetting $(AB)^{-1} = B^{-1}A^{-1}$ (order reversal), writing $A^{-1}B^{-1}$ instead.

12. GATE Traps

- $\text{tr}(AB) = \text{tr}(BA)$ even though $AB \neq BA$ — commonly tested trick.
- $(A + B)^{-1} \neq A^{-1} + B^{-1}$ — no such identity exists.

13. Memory Tricks "Reverse-and-invert / reverse-and-transpose" for $(AB)^{-1}$ and $(AB)^T$.

14. One-Page Summary Matrix = linear map representation. Key reversals: $(AB)^T = B^T A^T$, $(AB)^{-1} = B^{-1} A^{-1}$. Trace is cyclic-invariant. Invertible $\iff$ full rank $\iff$ nonzero det $\iff$ independent columns. Rank-Nullity: rank + nullity = number of columns.

5. Projection Matrix

1. Big Picture A projection matrix maps any vector onto a subspace (line, plane, column space) — the closest point in that subspace. Central to least squares and orthogonal decomposition.

2. Intuition Imagine shining a light perpendicular to a subspace and looking at the shadow of vector b on it — that shadow is Pb .

3. Formal Definition For a subspace with matrix A (columns = basis of subspace, full column rank), the projection matrix onto $C(A)$ is

$$P = A(A^T A)^{-1} A^T$$

satisfying $Pb \in C(A)$ and $(b - Pb) \perp C(A)$ for all b .

4. Explanation of Terms

- $C(A)$: column space of A , the subspace we project onto.
- $(A^T A)^{-1}$: exists only if A has full column rank (columns independent).
- Pb : the projection — closest point in $C(A)$ to b .
- $b - Pb$: residual/error vector, orthogonal to the subspace.

5. Important Properties

- $P^2 = P$ (idempotent) — *projecting an already-projected point changes nothing.*
- $P^T = P$ (symmetric) — **follows directly from the formula: $(A(A^T A)^{-1} A^T)^T = P$.**
- Eigenvalues of P are only 0 or 1 — **idempotency forces $\lambda^2 = \lambda$.**

- $\text{rank}(P) = \text{tr}(P) = \dim(C(A))$ — trace of idempotent = number of eigenvalue-1's = rank.
- $I - P$ is also a projection matrix (onto orthogonal complement) — $(I - P)^2 = I - 2P + P^2 = I - P$.

6. Theorems GATE Asks

- **Least Squares Solution:** $\hat{x} = (A^T A)^{-1} A^T b$ solves $\min \|Ax - b\|^2$; $A\hat{x} = Pb$ — apply whenever asked to fit a line/solve inconsistent $Ax = b$.
- Every symmetric idempotent matrix is a projection matrix onto its own column space — apply to identify if a given matrix is "a projection."

7. Important Formulae

- $P = A(A^T A)^{-1} A^T$
- Onto a single vector a : $P = \frac{aa^T}{a^T a}$
- $I - P$ projects onto orthogonal complement of $C(A)$.

8. Geometric Interpretation P flattens $\mathbb{R}^n$ onto the subspace $C(A)$ like a shadow on a wall; $I - P$ captures the perpendicular leftover.

9. Common Examples

- Projection onto a line through origin: $P = \frac{aa^T}{a^T a}$.
- Projection onto xy -plane in $\mathbb{R}^3$: $P = \text{diag}(1, 1, 0)$.

10. Common Non-Examples

- Any matrix that is idempotent but not symmetric is a projection (oblique), but NOT an **orthogonal** projection matrix.
- A rotation matrix (not idempotent) is never a projection matrix.

11. Common Mistakes Students Make

- Forgetting $(A^T A)^{-1}$ requires A to have independent columns.
- Confusing "projection matrix" (idempotent + symmetric) with just any idempotent matrix.

12. GATE Traps

- Asked "is P a projection matrix" — must check BOTH $P^2 = P$ AND $P^T = P$.
- Eigenvalues of a projection matrix are always 0/1 — used to quickly find rank via trace.

13. Memory Tricks "Project twice = project once" (idempotent), and "Projection never lies" (symmetric, real eigenvalues 0/1).

14. One-Page Summary $P = A(A^T A)^{-1} A^T$ projects onto $C(A)$; symmetric + idempotent ($P^2 = P = P^T$); eigenvalues only 0,1; rank = trace = dim of subspace; $I - P$ projects onto

orthogonal complement. Powers least-squares: $\hat{x} = (A^T A)^{-1} A^T b$.

6. Orthogonal Matrix

1. Big Picture Orthogonal matrices represent pure rotations/reflections — they preserve lengths and angles. Extremely useful because their inverse is just their transpose.

2. Intuition Columns are unit vectors, mutually perpendicular — like a rigid, non-distorting rotation of the coordinate axes.

3. Formal Definition A square matrix Q is orthogonal if

$$Q^T Q = Q Q^T = I \iff Q^{-1} = Q^T$$

4. Explanation of Terms

- $Q^T Q = I$: columns of Q are orthonormal (unit length, mutually perpendicular).
- $Q^{-1} = Q^T$: inverting an orthogonal matrix is free — just transpose.

5. Important Properties

- $\det(Q) = \pm 1$ — *since $\det(Q)^2 = \det(Q^T Q) = \det(I) = 1$.*
- Preserves length: $\|Qx\| = \|x\|$ — * $\|Qx\|^2 = x^T Q^T Q x = x^T x = \|x\|^2$.*
- Preserves inner product/angles: $(Qx) \cdot (Qy) = x \cdot y$ — *same identity extended to two vectors.*
- Product of orthogonal matrices is orthogonal — * $(Q_1 Q_2)^T (Q_1 Q_2) = Q_2^T Q_1^T Q_1 Q_2 = Q_2^T Q_2 = I$.*
- Eigenvalues have absolute value 1 (may be complex) — *length preservation forces $|\lambda| = 1$.*

6. Theorems GATE Asks

- $\det(Q) = +1 \Rightarrow$ pure rotation; $\det(Q) = -1 \Rightarrow$ reflection involved — apply when classifying a given orthogonal matrix geometrically.
- Orthogonal diagonalization: every **real symmetric** matrix $A = Q \Lambda Q^T$ with Q orthogonal — apply for symmetric matrix eigen-decomposition (spectral theorem).

7. Important Formulae

- $Q^{-1} = Q^T$
- $\det(Q) = \pm 1$
- 2D rotation matrix: $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

8. Geometric Interpretation Orthogonal transformations are rigid motions of space: rotations ($\det=+1$) and reflections ($\det=-1$); no stretching, no shearing.

9. Common Examples

- Identity matrix, rotation matrices, permutation matrices, reflection matrices (Householder).

10. Common Non-Examples

- Any matrix with non-unit-length columns.
- Symmetric matrix that isn't length-preserving (e.g. $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ — symmetric but not orthogonal).

11. Common Mistakes Students Make

- Confusing "orthogonal matrix" (special square matrix, $Q^T Q = I$) with "orthogonal vectors" (just perpendicular).
- Assuming any matrix with perpendicular columns (not unit length) is orthogonal — must be orthoNORMAL.

12. GATE Traps

- Non-square matrices can have orthonormal columns ($Q^T Q = I$) but are NOT called "orthogonal matrices" (that term is reserved for square).
- $\det(Q) = \pm 1$ does not by itself imply Q is orthogonal (converse false).

13. Memory Tricks "Orthogonal = its own inverse-transpose twin."

14. One-Page Summary $Q^T Q = Q Q^T = I \Rightarrow Q^{-1} = Q^T$. Preserves norms and angles; $\det Q = \pm 1$; eigenvalues have modulus 1. Used in spectral theorem: symmetric $A = Q \Lambda Q^T$.

7. Idempotent Matrix

1. Big Picture An idempotent matrix is one that "does its job once and repeating changes nothing" — the defining algebraic feature of projections.

2. Intuition Applying the transformation twice = applying it once. Once a vector lands in the target subspace, applying it again leaves it unchanged.

3. Formal Definition A square matrix A is idempotent if

$$A^2 = A$$

4. Explanation of Terms

- $A^2 = A$: applying A repeatedly stabilizes after the first application.

5. Important Properties

- Eigenvalues are only 0 or 1 — *if $Av = \lambda v$, then $A^2 v = \lambda^2 v = Av = \lambda v \Rightarrow \lambda^2 = \lambda$.*
- $\text{rank}(A) = \text{tr}(A)$ — trace = sum of eigenvalues = count of eigenvalue-1's = rank for diagonalizable idempotent.

- $I - A$ is also idempotent — $(I - A)^2 = I - 2A + A^2 = I - 2A + A = I - A$.*
- Only invertible idempotent matrix is the identity I — *if invertible, $A^2 = A \Rightarrow A = I$ by multiplying by A^{-1} .*
- Not necessarily symmetric (that extra condition gives orthogonal projection specifically).

6. Theorems GATE Asks

- If A is idempotent, A is diagonalizable with eigenvalues 0,1 only — apply to quickly compute powers ($A^k = A$ for all $k \geq 1$) or rank via trace.

7. Important Formulae

- $A^k = A$ for all $k \geq 1$.
- $\text{rank}(A) = \text{tr}(A)$.

8. Geometric Interpretation Represents a (possibly oblique) projection onto a subspace along some direction — symmetric idempotent = orthogonal projection; general idempotent = oblique projection.

9. Common Examples

- Any projection matrix $P = A(A^T A)^{-1} A^T$.
- $I, 0$ matrix (trivially idempotent).

10. Common Non-Examples

- Rotation matrices (not idempotent unless identity).
- Nilpotent matrices ($A^2 = 0 \neq A$ unless $A = 0$).

11. Common Mistakes Students Make

- Assuming idempotent implies symmetric (false — idempotent is weaker; symmetric+idempotent = orthogonal projection specifically).
- Forgetting only I is the invertible idempotent matrix.

12. GATE Traps

- Trick question: "Find rank of idempotent matrix given trace = k " $\rightarrow$ rank = k directly, no further computation.
- Confusing idempotent ($A^2 = A$) with involutory ($A^2 = I$) — different concepts entirely.

13. Memory Tricks "Idempotent = does it once, stays done." Eigenvalues only 0/1 — think "on/off switch."

14. One-Page Summary $A^2 = A$; eigenvalues $\in \{0, 1\}$; rank = trace; $I - A$ also idempotent; only invertible idempotent matrix is I . Symmetric idempotent = orthogonal projection matrix.

8. Partition Matrix and Their Properties

1. Big Picture Partitioning breaks a big matrix into smaller sub-blocks, letting you do matrix algebra (multiplication, inversion) block-by-block — very useful for structured/large matrices.

2. Intuition Same as normal matrix multiplication, but each "number" is itself a smaller matrix (block), and block-sizes must be compatible for multiplication just like scalar entries.

3. Formal Definition A matrix A is partitioned into blocks A_{ij} (submatrices) such that

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

and block operations (addition, multiplication) follow the same rules as scalar matrices provided block dimensions are **conformable** (compatible for multiplication).

4. Explanation of Terms

- **Block/submatrix:** a smaller matrix occupying a rectangular region of the original.
- **Conformable partitioning:** the column-partition of the first matrix must match the row-partition of the second matrix for block multiplication to work.

5. Important Properties

- Block multiplication: $\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & \cdots \\ \cdots & \cdots \end{pmatrix}$ — *works exactly like scalar 2×2 multiplication, entry-wise replaced by block matrix multiply (order of blocks matters!).*
- Block diagonal matrix determinant = $\prod \det(\text{diagonal blocks})$ — *diagonal blocks act independently, no cross terms.*
- Block triangular matrix determinant = product of determinants of diagonal blocks — *same reasoning as scalar triangular matrices, extended block-wise.*
- Inverse of block diagonal matrix = block diagonal matrix of inverses — *each block acts independently, so inverting block-wise suffices.*

6. Theorems GATE Asks

- **Schur complement formula** for block matrix inverse/determinant: for $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, $\det(M) = \det(A) \det(D - CA^{-1}B)$ (if A invertible) — apply when asked to compute determinant/inverse of a 2×2 block matrix.

7. Important Formulae

- $\det \begin{pmatrix} A & 0 \\ C & D \end{pmatrix} = \det(A) \det(D)$ (block triangular)
- Schur complement: $S = D - CA^{-1}B$; $\det(M) = \det(A) \det(S)$

8. Geometric Interpretation Not typically visualized geometrically; treat as bookkeeping tool for structured computation.

9. Common Examples

- Block diagonal matrices (independent sub-systems), block triangular matrices from Gaussian elimination on structured systems.

10. Common Non-Examples

- Randomly chopping a matrix into non-conformable blocks that can't be multiplied — not a valid partition for block operations.

11. Common Mistakes Students Make

- Multiplying blocks in wrong order (blocks don't commute, just like regular matrices) — must preserve $A_{ij}B_{jk}$ order strictly.
- Using scalar-like shortcuts (e.g. treating block matrix like a number) without checking conformability.

12. GATE Traps

- Determinant of block matrix is NOT simply $\det(A)\det(D) - \det(B)\det(C)$ (common wrong generalization of 2×2 scalar determinant formula) — only valid via Schur complement or if off-diagonal blocks are 0.
- Block matrix multiplication order errors due to non-commutativity of matrix blocks.

13. Memory Tricks "Blocks behave like numbers — but blocks don't commute!"

14. One-Page Summary Partition matrix into conformable blocks; treat block algebra like scalar algebra but respect non-commutativity. Block diagonal/triangular: $\det = \text{product of diagonal block dets}$. General 2×2 block $\det/\text{inverse}$ via Schur complement $D - CA^{-1}B$.

9. Quadratic Forms

1. Big Picture A quadratic form expresses a second-degree homogeneous expression in matrix form, letting you classify it (positive definite, indefinite, etc.) using eigenvalues.

2. Intuition It's a way to write expressions like $2x^2 + 3y^2 + 4xy$ using a symmetric matrix so you can study its "shape" (bowl, saddle, etc.) via eigenvalues.

3. Formal Definition For symmetric matrix A , the quadratic form is

$$Q(x) = x^T A x = \sum_i \sum_j a_{ij} x_i x_j$$

4. Explanation of Terms

- **Symmetric matrix A :** coefficients arranged so $a_{ij} = a_{ji}$; cross terms split evenly between a_{ij} and a_{ji} .
- $x^T A x$: scalar output, degree-2 homogeneous polynomial in $x_1, \dots, x_n$.

5. Important Properties

- Classification is determined entirely by the signs of eigenvalues of A — *since A is symmetric, it's orthogonally diagonalizable, and $Q(x)$ becomes $\sum \lambda_i y_i^2$ in new coordinates.*
- Positive definite $\iff$ all eigenvalues $> 0 \iff$ all leading principal minors > 0 — *equivalent tests via Sylvester's criterion.*
- Negative definite $\iff$ all eigenvalues < 0 .
- Indefinite $\iff$ eigenvalues have mixed signs.

6. Theorems GATE Asks

- **Sylvester's Criterion:** A positive definite $\iff$ all leading principal minors are positive — apply for quick definiteness check without computing eigenvalues.
- **Principal Axes Theorem:** $Q(x) = x^T A x$ can always be transformed via orthogonal substitution $x = Qy$ into $\sum \lambda_i y_i^2$ (no cross terms) — apply when asked to "reduce to canonical form."

7. Important Formulae

- $Q(x) = \sum_i \lambda_i y_i^2$ (in eigenbasis coordinates)
- Sylvester's test: check signs of leading principal minors $\Delta_1, \Delta_2, \dots, \Delta_n$.

8. Geometric Interpretation Positive definite $\rightarrow$ ellipsoid (bowl shape, min at origin); negative definite $\rightarrow$ inverted bowl; indefinite $\rightarrow$ saddle shape.

9. Common Examples

- $Q(x, y) = x^2 + y^2$ (positive definite, circle/sphere level sets).
- $Q(x, y) = x^2 - y^2$ (indefinite, saddle, hyperbola level sets).

10. Common Non-Examples

- Any form built from a non-symmetric matrix representation is ambiguous — always symmetrize first: $A_{sym} = \frac{A+A^T}{2}$.

11. Common Mistakes Students Make

- Using the given (possibly non-symmetric) matrix directly instead of symmetrizing it first.
- Confusing "positive semi-definite" (eigenvalues ≥ 0 , some can be 0) with "positive definite" (strictly > 0).

12. GATE Traps

- A matrix might look positive definite from diagonal entries alone (all positive) but fail Sylvester's test due to off-diagonal terms — must actually check minors/eigenvalues.
- Semi-definite matrices are singular ($\det = 0$) — definiteness at boundary.

13. Memory Tricks "Sign of eigenvalues = shape of the bowl." Positive→bowl-up, Negative→bowl-down, Mixed→saddle.

14. One-Page Summary $Q(x) = x^T A x$, A symmetric. Definiteness type = sign pattern of eigenvalues (equivalently, Sylvester's leading principal minors). Positive definite→all $\lambda_i > 0$ →ellipsoid; indefinite→saddle. Always symmetrize A first if given non-symmetric.

10. Systems of Linear Equations and Solutions; Gaussian Elimination

1. Big Picture Solving $Ax = b$ is the central computational problem of linear algebra — Gaussian elimination is the universal algorithm to find all solutions (unique, infinite, or none).

2. Intuition Row reduce the augmented matrix $[A|b]$ to echelon form; back-substitute. Whether a solution exists / is unique depends on comparing rank of A and rank of $[A|b]$.

3. Formal Definition Given $Ax = b$ (A is $m \times n$), **Gaussian elimination** transforms $[A|b]$ via elementary row operations to row echelon form (REF), then to reduced row echelon form (RREF), from which the solution set is read off.

4. Explanation of Terms

- **Augmented matrix** $[A|b]$: A with b appended as an extra column.
- **Elementary row operations**: (i) swap two rows, (ii) scale a row by nonzero constant, (iii) add multiple of one row to another — none change the solution set.
- **Pivot**: first nonzero entry in each row of echelon form.
- **Free variable**: a variable with no pivot in its column — can take any value.

5. Important Properties

- Elementary row operations don't change rank or solution set — *they're invertible operations, so solution sets before/after are identical.*
- Number of pivots = rank of the matrix — *definition of rank as number of independent rows/echelon leading entries.*
- Free variables = $n - \text{rank}(A)$ = nullity — *directly the count of unpivoted columns.*

6. Theorems GATE Asks

- **Existence & Uniqueness (Rouché–Capelli):**
 - No solution if $\text{rank}(A) < \text{rank}([A|b])$
 - Unique solution if $\text{rank}(A) = \text{rank}([A|b]) = n$
 - Infinite solutions if $\text{rank}(A) = \text{rank}([A|b]) < n$ — apply directly whenever asked to classify a system's solution type.

7. Important Formulae

- Number of free variables = $n - \text{rank}(A)$
- General solution = particular solution + null space solution: $x = x_p + x_h$

8. Geometric Interpretation Each equation is a hyperplane; solving the system = finding intersection of these hyperplanes (a point, line/plane, or empty set).

9. Common Examples

- Unique solution: two non-parallel lines intersecting at one point.
- Infinite solutions: two identical lines (same equation twice).
- No solution: two parallel, non-identical lines.

10. Common Non-Examples

- A homogeneous system $Ax = 0$ ALWAYS has at least the trivial solution — "no solution" is impossible for homogeneous systems.

11. Common Mistakes Students Make

- Forgetting to check $\text{rank}([A|b])$ separately from $\text{rank}(A)$ — a common source of wrong "unique solution" conclusions.
- Sign errors during row operations, especially row swaps affecting determinant sign (irrelevant for solving, but matters for det calculation).

12. GATE Traps

- Homogeneous system $Ax = 0$ with $m < n$ (more unknowns than equations) always has nontrivial (infinite) solutions.
- A square homogeneous system has only the trivial solution iff $\det(A) \neq 0$.

13. Memory Tricks "Compare ranks first: same+full $\rightarrow$ unique; same+less $\rightarrow$ infinite; different $\rightarrow$ none."

14. One-Page Summary Row reduce $[A|b]$ to echelon form. Compare $\text{rank}(A)$ vs $\text{rank}([A|b])$: unequal $\rightarrow$ no solution; equal & $= n \rightarrow$ unique; equal & $< n \rightarrow$ infinite (parametrize free variables). Homogeneous systems always have at least trivial solution; nontrivial solutions exist iff $\text{rank} < n$.

11. Eigenvalues and Eigenvectors

1. Big Picture Eigenvalues/eigenvectors reveal the "natural directions" of a matrix along which it acts as pure scaling — the single most GATE-tested concept in linear algebra.

2. Intuition An eigenvector is a direction that a matrix doesn't rotate — it only stretches/shrinks it (possibly flipping sign) by a factor called the eigenvalue.

3. Formal Definition For square matrix A , $v \neq 0$ is an eigenvector with eigenvalue λ if

$$Av = \lambda v$$

Eigenvalues are roots of the **characteristic equation**: $\det(A - \lambda I) = 0$.

4. Explanation of Terms

- **Characteristic polynomial:** $\det(A - \lambda I)$, degree- n polynomial in λ .
- **Eigenspace:** set of all eigenvectors for a given λ (plus zero vector) — a subspace.
- **Algebraic multiplicity:** number of times λ repeats as a root of characteristic polynomial.
- **Geometric multiplicity:** $\dim(\text{eigenspace}) = \text{nullity}(A - \lambda I)$.

5. Important Properties

- Sum of eigenvalues = $\text{trace}(A)$ — *coefficients of characteristic polynomial relate directly to trace.*
- Product of eigenvalues = $\det(A)$ — *characteristic polynomial constant term equals $\det(A)$ up to sign.*
- Geometric multiplicity $\leq$ algebraic multiplicity — *eigenspace dimension can't exceed root repetition.*
- Eigenvalues of triangular matrix = diagonal entries — *determinant of $(A - \lambda I)$ for triangular matrix is product of diagonal.*
- Eigenvalues of A^T = eigenvalues of A — * $\det(A - \lambda I) = \det((A - \lambda I)^T) = \det(A^T - \lambda I)$.*
- Eigenvalues of A^k are λ^k ; eigenvalues of A^{-1} are $1/\lambda$ (same eigenvectors) — *apply A repeatedly / invert the scaling relation.*
- Symmetric matrices have real eigenvalues and orthogonal eigenvectors (for distinct λ) — *spectral theorem consequence.*

6. Theorems GATE Asks

- **Cayley-Hamilton Theorem:** every matrix satisfies its own characteristic equation, $p(A) = 0$ — apply to compute high powers of A or find A^{-1} in terms of lower powers.
- **Diagonalization:** $A = PDP^{-1}$ possible iff A has n linearly independent eigenvectors — apply when asked to compute A^k efficiently or check diagonalizability.
- **Spectral Theorem:** real symmetric A is always orthogonally diagonalizable, $A = Q\Lambda Q^T$ — apply for symmetric matrices specifically (guaranteed diagonalizable).

7. Important Formulae

- $\det(A - \lambda I) = 0$
- $\sum \lambda_i = \text{tr}(A), \prod \lambda_i = \det(A)$
- $A^k = PD^kP^{-1}$ (if diagonalizable)

8. Geometric Interpretation Eigenvectors are the axes along which the linear transformation acts as pure stretching (no rotation); eigenvalue is the stretch factor along that axis.

9. Common Examples

- Diagonal matrix: eigenvalues = diagonal entries, eigenvectors = standard basis vectors.
- Identity matrix: eigenvalue 1 (multiplicity n), every nonzero vector is an eigenvector.

10. Common Non-Examples

- A rotation matrix in $\mathbb{R}^2$ (by angle $\neq 0, \pi$) has no real eigenvectors (eigenvalues are complex) — geometrically, rotation changes every real direction.

11. Common Mistakes Students Make

- Forgetting to check geometric multiplicity when determining diagonalizability (algebraic multiplicity alone isn't enough).
- Sign errors while expanding $\det(A - \lambda I)$.
- Assuming every matrix is diagonalizable (false — defective matrices like $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ aren't).

12. GATE Traps

- Repeated eigenvalue with geometric multiplicity $<$ algebraic multiplicity $\Rightarrow$ NOT diagonalizable (classic trap: shear-like matrices).
- A and A^T have same eigenvalues but generally DIFFERENT eigenvectors.
- Zero eigenvalue $\iff A$ is singular ($\det(A) = 0$).

13. Memory Tricks "Trace = sum, Det = product" of eigenvalues. "AM $\geq$ GM" for multiplicities (algebraic $\geq$ geometric) — mirrors the inequality name coincidentally, easy to remember direction.

14. One-Page Summary $Av = \lambda v$; solve $\det(A - \lambda I) = 0$. Sum of λ_i = trace, product = $\det(A)$. Diagonalizable iff geometric multiplicity = algebraic multiplicity for every eigenvalue. Symmetric matrices: always real eigenvalues, orthogonally diagonalizable. Cayley-Hamilton: A satisfies its own characteristic polynomial.

12. Determinant

1. Big Picture The determinant is a single scalar summarizing whether a matrix is invertible and how it scales volume/orientation — a gatekeeper value used across almost every topic.

2. Intuition $\det(A)$ = signed volume scaling factor of the unit cube under transformation A ; zero means the transformation squashes space into a lower dimension (non-invertible).

3. Formal Definition For $n \times n$ matrix A , using cofactor expansion along row i :

$$\det(A) = \sum_j (-1)^{i+j} a_{ij} M_{ij}$$

where M_{ij} is the minor (determinant of submatrix after deleting row i , column j).

4. Explanation of Terms

- **Minor** M_{ij} : determinant of the $(n - 1) \times (n - 1)$ submatrix obtained by deleting row i , column j .
- **Cofactor** $C_{ij} = (-1)^{i+j} M_{ij}$: signed minor.
- **Singular matrix**: $\det(A) = 0$, not invertible.

5. Important Properties

- $\det(AB) = \det(A) \det(B)$ — *determinant is multiplicative under composition of transformations.*
- $\det(A^T) = \det(A)$ — *cofactor expansion is symmetric under transpose.*
- Row swap flips sign of determinant — *from the alternating property of determinant under row exchange.*
- Scaling a row by k scales determinant by k — *linearity in each row.*
- Adding a multiple of one row to another doesn't change determinant — *used in Gaussian elimination-based determinant computation.*
- $\det(A) = \prod \lambda_i$ (product of eigenvalues) — *from characteristic polynomial constant term.*
- Triangular matrix determinant = product of diagonal entries — *cofactor expansion collapses to diagonal terms only.*

6. Theorems GATE Asks

- $\det(A) = 0 \iff A \text{ singular} \iff \text{columns dependent} \iff Ax = 0 \text{ has nontrivial solution}$ — apply for any "is this matrix invertible" question.
- **Cramer's Rule**: $x_i = \frac{\det(A_i)}{\det(A)}$ (replace i^{th} column with b) — apply for solving small (2×2 or 3×3) systems quickly, only when $\det(A) \neq 0$.

7. Important Formulae

- $\det(A^{-1}) = 1 / \det(A)$
- $\det(kA) = k^n \det(A)$ for $n \times n$ matrix
- $\det(A) = \prod \lambda_i$

8. Geometric Interpretation | $|\det(A)|$ = factor by which A scales area (2D) or volume (3D); sign indicates whether orientation is preserved (+) or flipped (-).

9. Common Examples

- $\det(I) = 1$; $\det$ of diagonal matrix = product of diagonal entries.
- 2×2 : $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.

10. Common Non-Examples

- Non-square matrices have NO determinant defined.

11. Common Mistakes Students Make

- Forgetting sign flip on row swaps when computing det via row reduction.
- Using $\det(A + B) = \det(A) + \det(B)$ (FALSE — no such identity in general).

12. GATE Traps

- $\det(kA) = k^n \det(A)$, NOT $k \det(A)$ — the exponent n is often missed.
- $\det(A + B) \neq \det(A) + \det(B)$ in general — classic wrong assumption tested.

13. Memory Tricks "Det is a volume detector — zero det means collapsed dimension."

14. One-Page Summary $\det(A) = 0 \iff$ singular. Multiplicative: $\det(AB) = \det(A) \det(B)$; $\det(A^T) = \det(A)$; $\det(kA) = k^n \det(A)$; equals product of eigenvalues. Row swap flips sign; row-scaling scales det; row addition leaves det unchanged (basis of elimination-based computation).

13. Rank

1. Big Picture Rank measures the "true dimensionality" of information contained in a matrix — how many independent rows/columns it truly has.

2. Intuition Rank = number of genuinely independent directions among the rows (or equivalently columns) of a matrix — the rest are redundant combinations.

3. Formal Definition

$\text{rank}(A) = \dim(C(A)) = \dim(\text{Row}(A)) =$ number of pivots in row echelon form of A

4. Explanation of Terms

- $C(A)$: column space, span of columns.
- $\text{Row}(A)$: row space, span of rows.
- **Pivot**: leading nonzero entry per row in echelon form.

5. Important Properties

- Row rank = Column rank always (even for non-square matrices) — *fundamental theorem of linear algebra, non-obvious but always true.*
- $\text{rank}(A) \leq \min(m, n)$ — *can't exceed number of rows or columns available.*
- $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$ — *composition can't increase independent directions.*
- $\text{rank}(A) = \text{rank}(A^T)$ — *direct consequence of row rank = column rank.*
- Elementary row/column operations don't change rank — *they are invertible, preserve the span dimension.*
- Full rank square matrix $\implies$ invertible — $\text{rank} = n \iff \det \neq 0$.

6. Theorems GATE Asks

- **Rank-Nullity Theorem:** $\text{rank}(A) + \text{nullity}(A) = n$ (columns) — apply whenever given one of rank/nullity, find the other.

- **Sylvester's Rank Inequality:** $\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB)$ — apply for bounding rank of a product.

7. Important Formulae

- $\text{rank}(A) \leq \min(m, n)$
- $\text{nullity}(A) = n - \text{rank}(A)$

8. Geometric Interpretation Rank = dimension of the image (output space) of the linear transformation A represents.

9. Common Examples

- Identity matrix: full rank = n .
- Zero matrix: rank = 0.
- Rank-1 matrix: $A = uv^T$ (outer product), all rows/columns are multiples of one vector.

10. Common Non-Examples

- A matrix with a row of all zeros always has rank $<$ number of rows.

11. Common Mistakes Students Make

- Computing rank by counting nonzero rows without fully reducing to echelon form first (a row can look "nonzero" but still be dependent before full reduction).
- Confusing rank of A with rank of $[A|b]$ when solving systems.

12. GATE Traps

- $\text{rank}(AB)$ can be LESS than both $\text{rank}(A)$ and $\text{rank}(B)$ — never assume equality.
- Rank of a matrix equals rank of its transpose — always, even for rectangular matrices (frequently tested as "surprising" fact).

13. Memory Tricks "Rank = number of pivots = number of truly independent rows/columns (same number either way!)"

14. One-Page Summary Rank = dim of column space = dim of row space = number of pivots. Row rank always equals column rank. rank + nullity = n . $\text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B)$. Full rank square matrix $\Rightarrow$ invertible.

14. Nullity

1. Big Picture Nullity counts the "wasted"/free directions of a matrix — dimensions that collapse to zero under the transformation. It's the direct complement of rank.

2. Intuition Nullity = number of independent solutions/directions you can move in and still land on zero when multiplied by A — i.e., how many free variables the system $Ax = 0$ has.

3. Formal Definition

$$\text{nullity}(A) = \dim(N(A)), \quad N(A) = \{x : Ax = 0\}$$

4. Explanation of Terms

- $N(A)$: null space (kernel), set of all vectors mapped to zero by A .

5. Important Properties

- $\text{nullity}(A) = n - \text{rank}(A)$ (Rank-Nullity Theorem) — *free variables = total columns - pivot columns*.
- $\text{nullity}(A) = 0 \iff$ columns of A are linearly independent $\iff Ax = 0$ has only trivial solution.
- For square A : $\text{nullity}(A) = 0 \iff A$ invertible — *combines rank-nullity with rank= n condition*.
- $N(A)$ is always a subspace — *closed under addition/scaling since $A(x + y) = Ax + Ay = 0$.*

6. Theorems GATE Asks

- **Rank-Nullity Theorem:** $\text{rank}(A) + \text{nullity}(A) = n$ — apply whenever given rank (or number of pivots), find nullity, or vice versa.

7. Important Formulae

- $\text{nullity}(A) = n - \text{rank}(A)$
- Number of free variables in $Ax = 0 = \text{nullity}$.

8. Geometric Interpretation Nullity = dimension of the subspace that gets "crushed" to the origin by the transformation A .

9. Common Examples

- Identity matrix: $\text{nullity} = 0$.
- Zero matrix ($n \times n$): $\text{nullity} = n$ (everything maps to zero).

10. Common Non-Examples

- A matrix with full column rank has nullity 0 — no non-example "trivial" nullspace beyond $\{0\}$ vector itself.

11. Common Mistakes Students Make

- Forgetting nullity is computed w.r.t. number of COLUMNS (n), not rows (m).
- Confusing nullity of A with nullity of A^T (generally different: $\text{nullity}(A^T) = m - \text{rank}(A)$).

12. GATE Traps

- For non-square $m \times n$ matrix with $m < n$: nullity is always $\geq n - m > 0$ — guaranteed nontrivial null space, even without computation.
- Nullity of A vs A^T are generally different unless A is square and full rank (both zero).

13. Memory Tricks "Rank + Nullity = Total columns — always adds up to n ."

14. One-Page Summary Nullity = $\dim(N(A)) = n - \text{rank}(A)$ = number of free variables in $Ax = 0$. Zero nullity $\Leftrightarrow$ independent columns $\Leftrightarrow$ (square case) invertible. For $m < n$ matrices, nullity is guaranteed positive.

15. Projections (General Concept)

1. Big Picture This is the general idea of "projection" — finding the component of one vector along a direction or subspace — before it's packaged into a projection matrix (Topic 5 covers the matrix form; this covers the underlying scalar/vector formulas).

2. Intuition Drop a perpendicular from vector b onto the direction/subspace of interest; the foot of that perpendicular is the projection.

3. Formal Definition Projection of vector b onto vector a (line through origin along a):

$$\text{proj}_a(b) = \frac{a^T b}{a^T a} a$$

Projection onto a subspace $C(A)$: $\hat{b} = A(A^T A)^{-1} A^T b$ (same as Topic 5's Pb).

4. Explanation of Terms

- $a^T b$: dot product, measures alignment between a and b .
- $\frac{a^T b}{a^T a}$: scalar coefficient — how much of a is "needed" to best approximate b .
- $b - \text{proj}_a(b)$: residual, orthogonal to a .

5. Important Properties

- Residual is always orthogonal to the projection direction/subspace — *by construction, minimizes distance, calculus/geometry gives perpendicularity*.
- Projecting onto a subspace and then projecting again onto the same subspace changes nothing — *reflects idempotency of the projection matrix*.
- Projection minimizes $\|b - x\|$ over all x in the target subspace — *closest-point property, foundation of least squares*.

6. Theorems GATE Asks

- **Best Approximation Theorem:** the projection $\hat{b}$ onto subspace W is the unique closest point in W to b — apply for least-squares/approximation questions.
- **Gram-Schmidt Process:** repeatedly subtract projections onto previous vectors to construct an orthogonal/orthonormal basis — apply whenever asked to orthogonalize a given basis.

7. Important Formulae

- Scalar projection (length only): $\frac{a^T b}{\|a\|}$
- Vector projection onto a : $\frac{a^T b}{a^T a} a$

- Gram-Schmidt step: $u_k = v_k - \sum_{i < k} \text{proj}_{u_i}(v_k)$

8. Geometric Interpretation The shadow of b falling directly (perpendicularly) onto the line/plane/subspace defined by a or A .

9. Common Examples

- Projecting $b = (3, 4)$ onto $a = (1, 0)$ gives $(3, 0)$.
- Gram-Schmidt on 2 independent vectors in $\mathbb{R}^2$ yields an orthogonal pair.

10. Common Non-Examples

- Projecting onto the zero vector is undefined (division by $a^T a = 0$).

11. Common Mistakes Students Make

- Confusing scalar projection (a number, length) with vector projection (a vector).
- Forgetting to normalize by $a^T a$ (using $a^T b \cdot a$ instead of $\frac{a^T b}{a^T a} a$).

12. GATE Traps

- Gram-Schmidt order matters — projections must be subtracted using the ALREADY orthogonalized vectors, not the original ones, at each step.
- Vector projection formula requires $a \neq 0$; watch for such degenerate cases in questions.

13. Memory Tricks "Projection coefficient = (how aligned)/(how big the target is)" = $\frac{a^T b}{a^T a}$.

14. One-Page Summary $\text{proj}_a(b) = \frac{a^T b}{a^T a} a$; onto subspace: $\hat{b} = Pb$ (Topic 5's projection matrix). Residual $b - \hat{b}$ is orthogonal to target. Basis for Gram-Schmidt orthogonalization and least-squares approximation.

16. LU Decomposition

1. Big Picture LU decomposition packages the steps of Gaussian elimination into two triangular matrices, making repeated solving of $Ax = b$ (for different b 's) much faster.

2. Intuition L records the row-reduction multipliers used to eliminate entries below the pivots; U is the resulting upper triangular (echelon) matrix. Multiplying them back reconstructs A .

3. Formal Definition For a square matrix A (that doesn't require row swaps), there exist matrices such that

$$A = LU$$

where L is lower triangular with 1's on diagonal (unit lower triangular), and U is upper triangular (row echelon form of A).

4. Explanation of Terms

- L : lower triangular matrix storing the elimination multipliers used to zero out below-diagonal entries.
- U : upper triangular matrix — the result after forward elimination.
- $PA = LU$: when row swaps ARE needed, a permutation matrix P is introduced.

5. Important Properties

- Exists (without permutation) iff all leading principal minors of A are nonzero — *elimination can proceed without needing row swaps only if each pivot step is valid.*
- $\det(A) = \det(L) \det(U) = \prod (\text{diagonal of } U)$ (since $\det(L) = 1$) — *triangular matrix determinants are just diagonal products.*
- Once $A = LU$ is known, solving $Ax = b$ reduces to two easy triangular solves: $Ly = b$ (forward substitution), then $Ux = y$ (back substitution) — *much cheaper than re-doing elimination for each new b .*

6. Theorems GATE Asks

- Solving $Ax = b$ via LU is $O(n^2)$ per new b (after one-time $O(n^3)$ decomposition) — apply when comparing efficiency of solving multiple systems with same A , different b .
- LU decomposition is **not unique** without a normalization convention (e.g., unit diagonal in L) — apply when asked to verify/construct LU factorization.

7. Important Formulae

- $A = LU$ (or $PA = LU$ with row swaps)
- $\det(A) = \prod u_{ii}$ (diagonal entries of U)
- Solve via: $Ly = b$, then $Ux = y$

8. Geometric Interpretation Not typically visualized geometrically — it's a computational/algebraic tool for efficient equation-solving.

9. Common Examples

- Any matrix reducible without row swaps, e.g. $\begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$.

10. Common Non-Examples

- A matrix requiring a row swap during elimination (e.g., zero pivot) has NO plain LU factorization — needs $PA = LU$ instead.

11. Common Mistakes Students Make

- Forgetting the sign convention: multipliers in L are the **NEGATIVE** of what you added during elimination (since you're "undoing" the elimination step).
- Assuming LU decomposition is always unique — it isn't, without a normalization like unit diagonal in L .

12. GATE Traps

- If a zero pivot appears mid-elimination, plain LU fails — must use $PA = LU$; this is a common "why doesn't LU exist here" trap.
- Determinant via LU: don't forget $\det(P) = \pm 1$ contributes a sign if row swaps were used.

13. Memory Tricks "L = Lower (multipliers), U = Upper (echelon result)." Think "Lower keeps the recipe, Upper is the finished dish."

14. One-Page Summary $A = LU$: L unit-lower-triangular (multipliers), U upper triangular (echelon form). Exists without swaps iff leading principal minors nonzero; else use $PA = LU$. Solve $Ax = b$ via forward substitution ($Ly = b$) then back substitution ($Ux = y$). $\det(A) = \prod(\text{diag of } U)$.

17. Singular Value Decomposition (SVD)

1. Big Picture SVD is the most general matrix factorization — it works for ANY matrix (even non-square, singular) and reveals its true geometric action: rotate, stretch, rotate again.

2. Intuition Every linear transformation, no matter how weird, can be broken into: rotate input space (V), stretch along orthogonal axes (Σ), rotate output space (U). SVD finds these three universal ingredients.

3. Formal Definition For any $m \times n$ matrix A , there exist orthogonal matrices U ($m \times m$), V ($n \times n$), and a diagonal-like matrix Σ ($m \times n$, nonnegative entries on diagonal, rest zero) such that

$$A = U\Sigma V^T$$

4. Explanation of Terms

- U : orthogonal matrix whose columns are eigenvectors of AA^T (left singular vectors).
- V : orthogonal matrix whose columns are eigenvectors of $A^T A$ (right singular vectors).
- Σ : diagonal matrix of **singular values** $\sigma_i = \sqrt{\lambda_i(A^T A)}$, arranged in decreasing order.
- **Singular values**: nonnegative square roots of eigenvalues of $A^T A$ (equivalently AA^T).

5. Important Properties

- $A^T A = V\Sigma^2 V^T$ and $AA^T = U\Sigma^2 U^T$ — *substituting $A = U\Sigma V^T$ and using $U^T U = V^T V = I$.*
- Number of nonzero singular values = $\text{rank}(A)$ — *singular values are square roots of nonzero eigenvalues of $A^T A$, which count independent directions.*
- SVD exists for EVERY matrix (rectangular, singular, anything) — unlike eigendecomposition which needs square diagonalizable matrices.

- $\|A\|_2 = \sigma_{max}$ (largest singular value) — *operator norm equals the largest stretch factor.*
- For symmetric positive semi-definite A : singular values = eigenvalues (SVD coincides with eigendecomposition).

6. Theorems GATE Asks

- **Eckart-Young Theorem:** best rank- k approximation of A (minimizing Frobenius/operator norm error) is obtained by keeping only the top k singular values/vectors — apply for low-rank approximation / compression questions.
- SVD always exists (universality theorem) — apply whenever asked to decompose a non-square or singular matrix (where eigendecomposition fails).

7. Important Formulae

- $A = U\Sigma V^T = \sum_{i=1}^r \sigma_i u_i v_i^T$ (sum of rank-1 terms, $r = \text{rank}(A)$)
- $\sigma_i = \sqrt{\lambda_i(A^T A)}$
- $\text{rank}(A) = \text{number of nonzero } \sigma_i$

8. Geometric Interpretation V^T rotates input space to align with principal axes; Σ stretches along those axes by σ_i ; U rotates the result into output space. A unit sphere maps to an ellipsoid with semi-axes σ_i .

9. Common Examples

- SVD of identity matrix: $U = V = I, \Sigma = I$.
- SVD used in PCA: singular values of centered data matrix relate directly to variance captured by principal components.

10. Common Non-Examples

- Eigendecomposition ($A = PDP^{-1}$) is NOT the same as SVD in general — eigendecomposition requires square diagonalizable A , and P need not be orthogonal.

11. Common Mistakes Students Make

- Confusing SVD ($A = U\Sigma V^T$, always exists) with eigendecomposition ($A = PDP^{-1}$, needs square + diagonalizable).
- Forgetting singular values are always non-negative (unlike eigenvalues, which can be negative or complex).

12. GATE Traps

- For a symmetric matrix with a negative eigenvalue λ , the corresponding singular value is $|\lambda|$, NOT λ — sign gets absorbed into U or V .
- Rank of A = number of NONZERO singular values, not total count of σ_i slots in Σ .

13. Memory Tricks "Rotate-Stretch-Rotate" (V rotates in, Σ stretches, U rotates out).

14. One-Page Summary $A = U\Sigma V^T$: U, V orthogonal, Σ has singular values $\sigma_i = \sqrt{\lambda_i(A^T A)} \geq 0$ on diagonal. Exists for ANY matrix. $\text{rank}(A)$ = number of nonzero σ_i . Best rank- k approximation = keep top k singular values (Eckart-Young). Distinct from eigendecomposition, which needs square diagonalizable matrices.

End of Notes — 17 Topics Covered.